System Design Workshop: Real-time Recommendation System

Scaler School of Technology | ML System Design and LLMOps | Week 5 — Session 1/2 | Student Worksheet

The ML System Design Rubric — Your Framework for Today

Use this 10-step framework to design the system. Work through the steps in order on pages 2–3.

1. **Clarify the Problem** — scope, constraints, success metrics
2. **Context** — business objective, users, what failure costs
3. **Problem Statement** — input, output, what to predict/rank
4. **Define Requirements** — functional and non-functional
5. **Estimations** — users, QPS, data volume, storage, compute
6. **System Architecture** — (a) Data Collection • (b) Data Engineering • (c) Feature Engineering • (d) Preprocessing • (e) Models • (f) Post-processing
7. **Evaluation** — offline metrics, online north-star, guardrails
8. **A/B Testing** — unit, split, duration, biases to flag
9. **Deployment & Monitoring** — pipelines, rollout, dashboards
10. **Risk & Trade-offs** — latency vs. accuracy, ethical risks, failure modes

The Problem

Design a real-time music recommendation system for a streaming service with **50 million active users**.

Requirements: personalised recommendations • p99 latency $<$ 100ms • 10,000 RPS peak • new songs visible within 24h • cold-start handling for new users

Steps 1–3 — Clarify, Context, Problem Statement

Ask your clarifying questions, note the context, then write a precise ML problem statement.

Clarifying questions I would ask:

Context (business objective, users, cost of failure):

Problem statement (input, output, constraint):

Step 4 — Requirements

Requirement	Value / assumption	Implication for design
Latency SLA		
Throughput (peak RPS)		
Availability		
Freshness SLA		
Cold-start		

Step 5 — Estimations

What	Your estimate	Design implication
Daily event volume		
User feature store size		
Song embedding index size		
Serving pods needed		

Step 6 — System Architecture

Label each component: what it does, key tools, key decisions. Draw the full system diagram in the box.

(a–c) Data Collection, Engineering & Feature Store

(d–e) Preprocessing / Candidate Gen. & Models

(f) Post-processing & Full System Diagram

Draw components as labelled boxes. Show data flow with arrows. Label every arrow.

Step 7 — Evaluation

Type	Metric(s)	Why this metric?
Offline		
Online north-star		
Guardrail		

Step 8 — A/B Testing

Unit of randomisation, split, duration, sample size, and one bias you would flag.

Step 9 — Deployment and Monitoring

Deployment (pipeline, rollout, registry):

6 metrics to monitor:

1.
2.
3.
4.
5.
6.

Step 10 — Risk Management and Trade-offs

Trade-off	Your choice	Reasoning
Pre-computed vs. real-time candidates		
Two-stage vs. single-ranker		
Implicit vs. explicit feedback		
Cold-start strategy		
Diversity vs. CTR		

What Breaks First at 10× Load?

Identify the bottleneck and explain how you would fix it.

Name: _____ Group: _____ Date: _____